Case Report

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The Action of Prostaglandins on Ciliary Hypertrichosis: A Case Report of Pachydermoperiostosis

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ABSTRACT

Pachydermoperiostosis or primary hypertropic osteoarthropathy is a rare hereditary disorder that was first described in 1868. It is characterized by digital clubbing, pachydermia (thickening of the facial skin and/or scalp), and periostosis (swelling of periarticular tissue and subperiosteal new bone formation). We report a case of a patient with the complete form of the disease, and with a unique appearance of the hair shaft and eyelashes. The authors propose a possible mechanism to justify the abnormalities observed in the patient's hair shafts regarding the metabolism of prostaglandins and its relationship with the hair follicle physiological cycle.

Key words: Cutis verticis gyrata, eyelashes, hair, pachydermoperiostosis, primary hypertrophic osteoarthropathy, prostaglandins

INTRODUCTION

Pachydermoperiostosis or primary hypertropic osteoarthropathy is a rare hereditary disorder that was first described in 1868. It is characterized by digital clubbing, pachydermia and periostosis.[1,2]

Touraine *et al*. described three forms of pachydermoperiostosis: (1) A complete form with pachydermia and periostosis, (2) an incomplete form with evidence of bone abnormalities but lacking pachydermia, and (3) a forme frusta with prominent pachydermia and minimal-to-absent skeletal changes.[3,4]

We describe a case of a patient with the complete form of the syndrome, focusing on the changes in the hair and eyelashes. The authors suggest a possible mechanism for the observed changes.

CASE REPORT

A 22-year-old male patient, reports 5 years ago the initial development of facial skin thickening with accentuation of the ruts, pustules and cystic lesions on the face, and palmoplantar hyperhidrosis.

Physical examination revealed thickening of facial skin with accentuated skin ruts and circumvolutions mainly in the frontal region and scalp (*cutis verticis gyrata*), presence of dense and voluminous hair, keloid scars, hypertrophic eyelid skin, and lengthened eyelashes [Figures 1 and 2]. The joint examination revealed digital clubbing of the hands and feet, and knees with swelling and joint effusion.

He had radiographs showing diffuse periosteal thickening. Laboratorial exams were normal.

Based on clinical and radiological findings, age of onset of the disease, and absence of underlying illnesses, the diagnosis of the full form of pachydermoperiostosis was established.

DISCUSSION

The pathogenesis of pachydermoperiostosis involves the Human Gene and Protein Database mutation, gene that encodes the 15-hydroxyprostaglandin dehydrogenase, a key enzyme responsible for the degradation of prostaglandins (PGs).[5,6] Individuals with this mutation show high levels of PGs, especially PG E2 (PGE2). The clinical manifestations are due to the excessive formation of collagen by fibroblast hyperactivation promoted by PGE2.

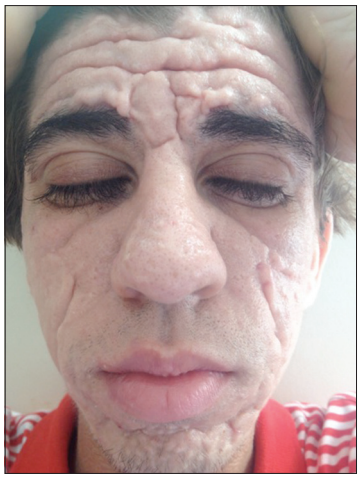


Figure 1: Thickening of facial skin with accentuated skin ruts and circumvolutions mainly in the frontal region, keloid scars, hypertrophic eyelid skin and lengthened eyelashes

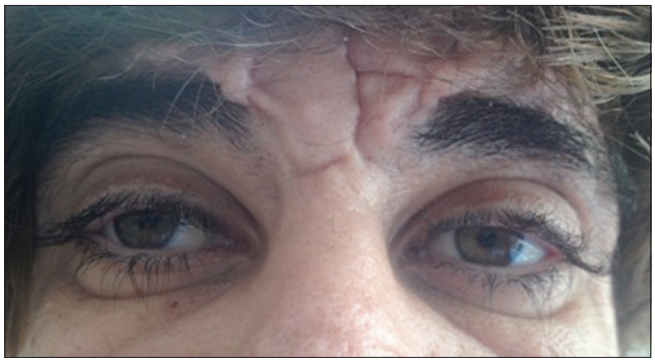


Figure 2: Hypertrophic eyelid skin and lengthened eyelashes

We believe that, in this case, the changes of the scalp hair and eyelashes can also be attributed to high levels of PG.

Side-effects as cilliary hypertrichosis and eyelid hypertrophy, were observed in patients with topical PG analogues for glaucoma treatment.[7] This encouraged studies about their action in the hair follicle, and are currently approved by the Food and Drug Administration for the treatment of alopecia and cilliary hypotrichosis. The mechanism by which these drugs induce hypertrichosis is not totally elucidated, but experimental studies with rats have shown how they interfere in the hair cycle.[8,9]

Receptors for PGE2 and F2α are described in hair follicles, especially in the follicular papilla.[10] They induce telogen follicles to enter the anagen phase early, in addition to promoting an extension of the same, delaying the onset of catagen phase. Other effects consist in follicular melanogenesis and thickening of the hair bulb, which leads to hyperpigmentation and hair thickening.[8,9] Thus longer, pigmented and thicker hair would have been noticed, which could explain the findings in this patient.

We believe that in this case, the abnormalities found in the hair of the scalp and eyelashes can be attributed to high levels of prostaglandins present in patients with pachydermoperiostosis.

There are no studies to corroborate that this mechanism can be applied to the hair of the scalp. However the clinical findings in this case favor this hypothesis.

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